UNIT - 12

TPS

Three Phase Supply

(Poly Phase System)

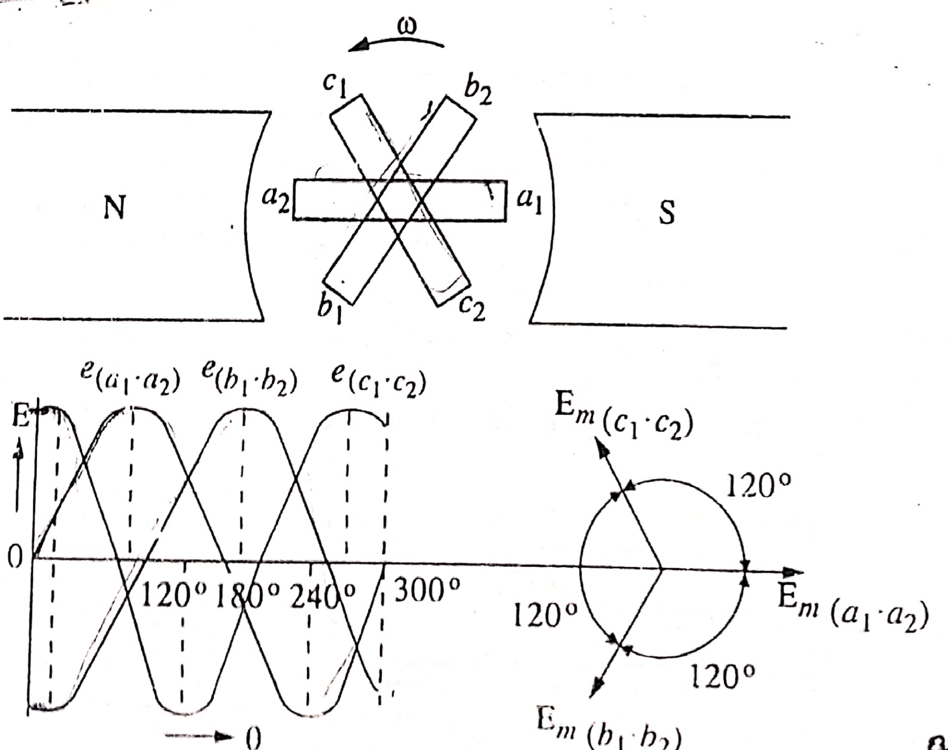
armature winding

Introduction. The alternating currents and voltages discussed so far are known as single phase voltages and currents. Single phase circuit consists of single alternating voltage and current wave. The generator which produces a single phase supply (known as single phase alternator) has only one armature winding. But if the number of armature windings are increased to two or three, it is known as 2–phase generator and three phase generator respectively. These windings are equally displaced from one another, the value of the angle is determined by the number of windings or phases.

In 2-phase generator the armature windings are displaced by 90 electrical degrees apart and in 3-phase generator, there are three identical windings which are 120° electrical apart from each other. A three phase generator therefore, produces three voltages of the same magnitude and direction.

TPS–1. POLY–PHASE SYSTEM

A ploy–phase generator has two or more armature windings equally displaced from each other. These windings are acted upon by the common magnetic field and thus each winding or phase produces single alternating voltage of the same magnitude and frequency.



Baambir Sidhu

Fig. TPS–1

Fig. TPS–1 shows a three phase generator. It has three windings A, B and C splaced by 120° $\left(= \frac{360 \text{ Electrical degree}}{\text{No. of phases}}\right)$.

The three coils are rotating in anti-clock-wise direction with an angular velocity ω . Here the ends a_1, b_1, and c_1 are 120° apart from each other. Since all the three oils are similar and also are rotating with the same angular velocity, hence *e.m.f.* nduced in them will be of the same magnitude and frequency, but the phase difference etween them will be 120° as shown in the wave diagram. The *e.m.f.* in coil B will e 120° lagging that of coil A and *e.m.f.* in coil C will be 240° lagging that of coil . The equations of these *e.m.f.* are as under.

$$e_{a_1a_2} = E_{max} \sin \omega t \quad - R$$

$$e_{b_1b_2} = E_{max} \sin (\omega t - 120°) \quad - Y$$

$$e_{c_1c_2} = E_{max} \sin (\omega t - 240°)$$

Naming the phases :

The three phases or windings may be numbered as 1, 2, 3, or A, B, C. But it is customary that they may be given natural colours. The colours used are Red (R), ellow (Y) and Blue (B). Hence in that case the sequence is R Y B. Now in 3-phase stem, there can be two possible sequences in which the three coils or phase oltages may pass through their maximum value *i.e.*, Red → Yellow → Blue or Red → Blue → Yellow. By convention, R Y B is taken as positive and R B Y as egative.

Double Script Notation. In the double script notation, two letters are written the foot of the symbol for voltage or current. This conveys the following formations ;

—The two letters indicate the two points between which voltage or current ists and the order of the letters indicate the positive direction of the quantity in hich it acts. For example, if the voltage is represented by V_{RY}, then it indicates oltage between points R and Y and with point R positive *w.r.t.* point Y during its ositive half cycle. Similarly I_{RY} indicates current I between points R and Y and its irection from R to Y during its positive half cycle.

—The double script notation is very useful in solving problems on *a.c.* circuits ving a number of voltages and currents.

PS-2. ADVANTAGES OF THREE PHASE SYSTEM OVER SINGLE PHASE SYSTEM

Power delivered is constant

The advantages of three phase system over single phase system are given elow :

1 In a single phase circuit the power delivered is pulsating. Even if current and voltage are in phase, the power is zero twice in each cycle, and when the current leads or lags behind the voltages, the power is negative twice and zero four times during each cycle. This is not objectionable for lighting and small motors but with large motors it causes excessive vibrations. In three phase system the total power delivered is constant if loads are balanced though the power of any one phase or ... three phase system is highly desirable particularly fo...

2. The rating of a given machine increases with the increase in number of phases.

3. Single phase induction motors are not self start, so they need some auxiliary means of starting, but in case of three-phase induction motors, there is no need of providing any auxiliary means for starting.

4. Power factor of single-phase motor is generally lower than that of a three phase motor of the same rating. The efficiency of three phase motor is more than that of single phase motor.

5. Three phase system requires $\frac{3}{4}$th weight of copper of transmission lines than that required by single phase system to transmit the same amount of power at a particular voltage and over the same distance.

6. Rotating magnetic field can be produced by passing three phase current through stationary coils.

7. Three phase system is more efficient and reliable than single phase system.

8. Parallel operation of three phase alternators is simple as compared to that of single phase alternators because of pulsating reaction in single phase alternators.

TPS-3. PHASE SEQUENCE

In three phase system, there are three equal phase voltages which are displaced by an angle of 120° electrical as shown in fig. TPS-1. It is very important to know the order in which the phase voltages reach their maximum positive values. This is called **phase sequence**.

Hence, **phase sequence** is defined as, *the order in which the voltages (e.m.f. s) in the three phases reach their maximum values.*

In fig. TPS-1. The *e.m.f.* s in the three phases attain their positive maximum value in the order of a_1a_2, b_1b_2 and c_1c_2, therefore their phase sequence is *a, b* and *c* respectively. However if the names of the three coils are as Red, Yellow and Blue, then their phase sequence will be R, Y, B.

The knowledge of phase sequence is very essential ;

(i) For parallel operation of 3-phase alternators and transformers.

(ii) The direction of rotation of 3-phase induction motor also depends upon the phase sequence of the 3-phase supply. If the direction of rotation of motor is to be reversed. Then reverse the phase sequence of the supply.

Types of loads

There can be two types of 3-phase loads which are as under ;

(i) Star connected load

(ii) Delta connected load.

If Z_1, Z_2 and Z_3 be the three loads (impedences) connected in star or delta as shown in fig. TPS-2 are equal in magnitude and phase angle, then the 3-phase load is said to be a balanced load. Under such conditions, all the phase or line currents and all phase and line voltages are equal in magnitude

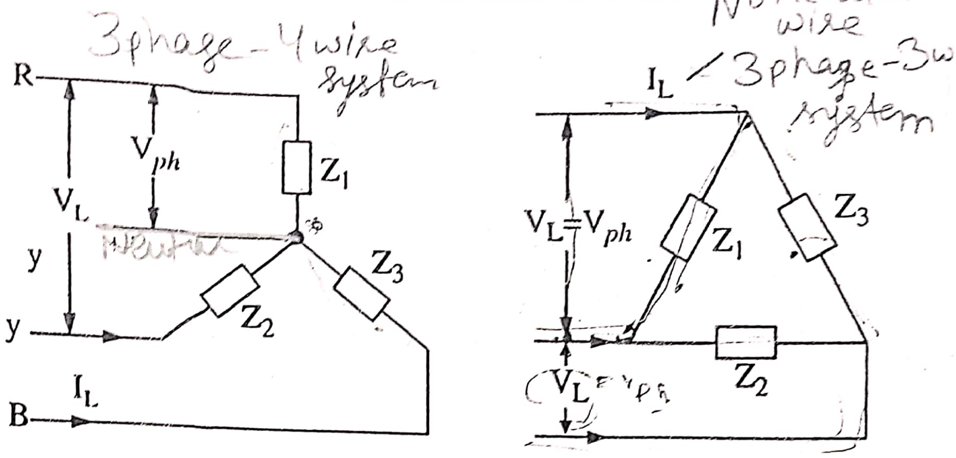


Fig. TPS-2

The 3-phase load can be balanced and unbalanced, however in this book 3-phase balanced load is taken unless stated, otherwise.

TPS–4. INTER CONNECTIONS OF THREE PHASES

In a three phase generator (alternator), there are three armature windings or phases. Each winding has two terminals (Start and Finish). The total no. of conductors in that case, being six. Each phase supplies an independent load across a pair of wires or leads as shown in Fig. TPS-3. This will make the whole system complicated and expensive. Hence the three phases are generally inter connected, which results in the saving of copper. The two universally adopted methods are :

1. Star or Wye (Y) connections.
2. Mesh or Delta (Δ) connections.

In star
emf b/w any one line and neutral gives phase emf
emf b/w two given lines gives line emf.

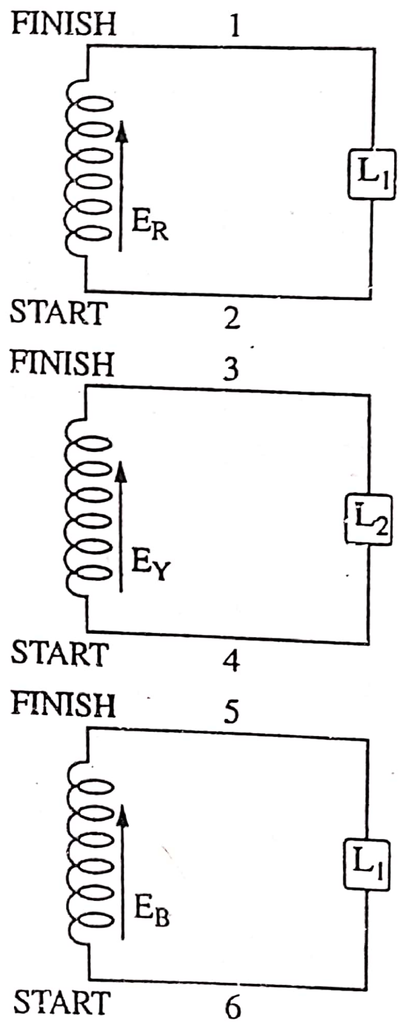
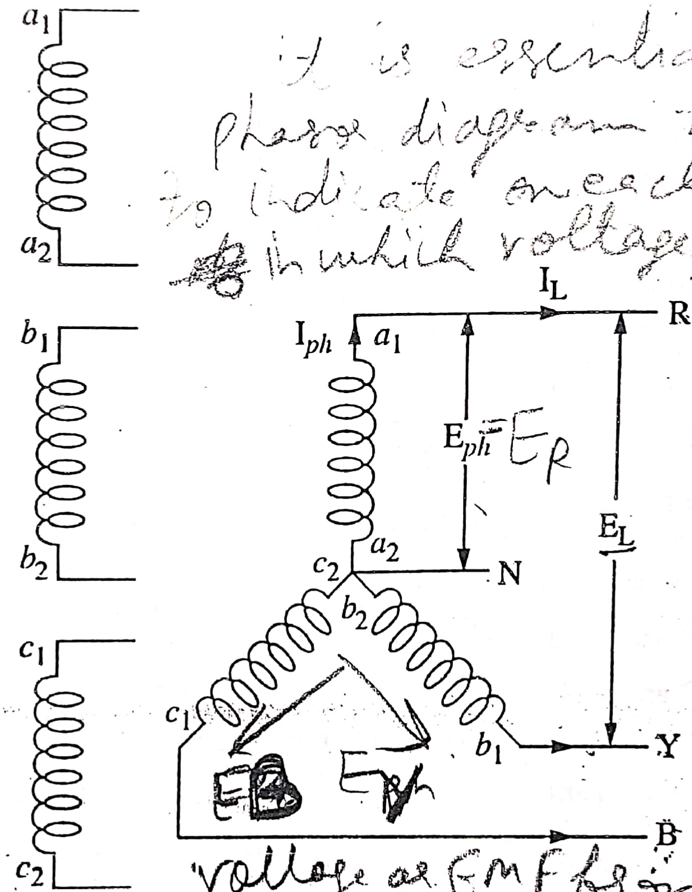


Fig. TPS-3

TPS–5. STAR OR Y-CONNECTIONS

In this method of inter connections, the similar ends (Either start or finish) of three phases are connected together to a common point known as star or neutral point N. The three line conductors run from the three ends and are named as R, Y and B.

The figure TPS–4 shows star connections. The voltage between any phase wire and neutral is called phase voltage and is represented as E_{ph} and the voltage between any two phases or lines is called line voltage and is represented by E_L.



it is essential to relate phasor diagram to ckt dia. and to indicate on each phase direction in which voltage or current is assumed to be +ve.

Assume emf in each phase +ve when acting from neutral point outwards

voltage or EMF from R via N to Y is phasor difference of E_R and E_Y

Fig. TPS–4

The current in each phase is called phase current and is written as I_{ph} and the current in each line is called current and is written as I_L.

TPS–6. RELATIONSHIP BETWEEN LINE AND PHASE VOLTAGES AND CURRENT IN A STAR CONNECTED SYSTEM :

sides of ||gram are equal diagonals bisect one another at right angles

Phase Voltage. Voltage measured between the two own terminals of the coil is known as **Phase voltage** and the current flowing is known as **Phase current.**

Line Voltage. Voltage measured between the two outer terminals of the two coils as shown in fig. TPS–5 *i.e.*, a cross R, Y is known as line voltage and the current flowing in the main leads is known as line current.

Let us assume the *e.m.f.* in each phase to be positive when acting from the neutral point outwards so that the *r.m.s.* values

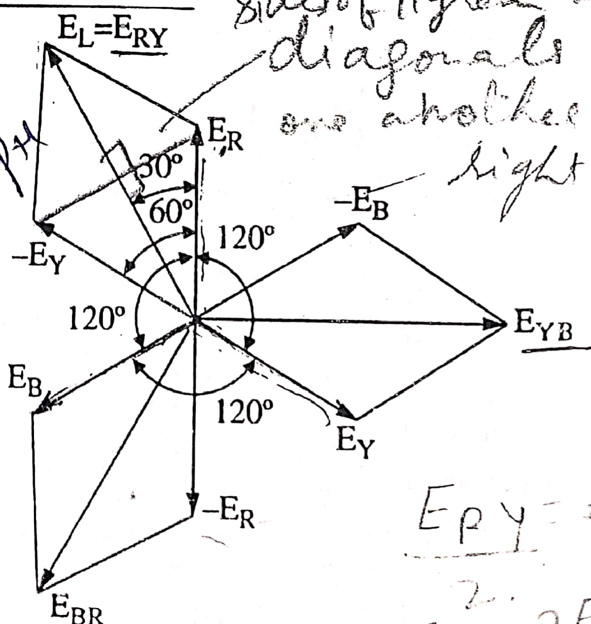


Fig. TPS-5

$E_{RY} = E_R ...$

$E_{RY} = 2E_R \cos 30$

of the *e.m.f.* generated in the three phases can be represented by E_R, E_Y and E_B as shown in figure TPS–5. The value of *e.m.f.* acting from R to Y (through star point) is the vector difference of E_R and E_Y.

From vector diagram shown in figure TPS–5.

Reversing E_Y, therefore adding the voltage,

$$E_{RY} = E_L = E_R + (-E_Y) \text{.... vectorially}$$

or $$E_L = E_R - E_Y$$

Let, $$E_R = E_Y = E_B = E_{ph} \text{ (phase voltage)}$$

and $$E_{RY} = E_{YB} = E_{BR} = E_L \text{ (Line voltage)}$$

The phase angle between E_R and $(-E_Y) = 60°$

$\therefore$ $$E_L^2 = E_R^2 + E_Y^2 + 2\,E_R.\,E_Y.\cos 60$$

or $$E_L^2 = 3\,E_{ph}^2 \quad \text{or} \quad E_L = \sqrt{3}\,.\,E_{ph}$$

Hence, $$E_L = \sqrt{3}\,.\,E_{ph} \text{ in star connections.}$$

Thus in star connections as seen from the figure,

(i) Line voltage $= \sqrt{3}$. Phase voltage.

(ii) Line voltages are 120° apart.

(iii) Line voltages are 30° ahead of their respective voltages.

Line currents and Phase currents. Each line is in series with its individual phase winding hence the line current in each is shown in fig. TPS–6.

Current in red line $= I_R$

Current in yellow line $= I_Y$

Current in blue line $= I_B$

Assuming balanced load *i.e.*,

$$I_R = I_Y = I_B$$
$$= I_{ph} \text{ (say) phase current}$$

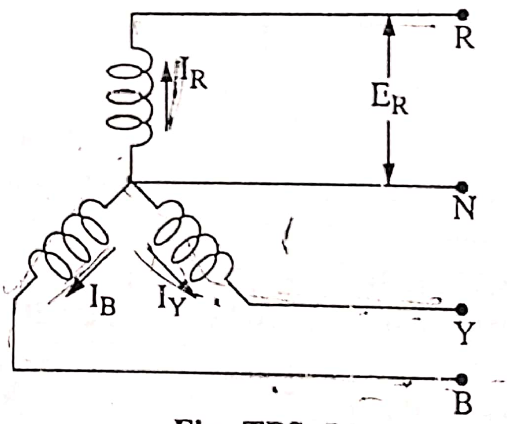


Fig. TPS–5

Then line current, $I_L = I_{ph}$

Power. Total power in the star connections is the sum of the three phase powers *i.e.*,

Total Power $= 3 \times$ phase power.

Let, E_R = R.M.S. value of E.M.F. per phase

I_R = R.M.S. value of current per phase

and $\cos\phi$ = the power factor

$\therefore$ Power per phase $= E_R.\,I_R \cos\phi = E_{ph}.I_{ph}.\cos\phi$

Total Power in 3 phase $= 3 \times E_{ph} \times I_{ph} \times \cos\phi$

$$E_{ph} = \frac{E_L}{\sqrt{3}} \quad \text{and} \quad I_{ph} = I_L$$

$$\therefore \quad \text{Power} = 3 \times \frac{E_L}{\sqrt{3}} \cdot I_L \cos\phi$$

or

$$P = \sqrt{3} \cdot E_L \cdot I_L \cdot \cos\phi$$

Here ϕ is the angle between phase voltage and phase current.

Some important relations is star connections :

(i) Line voltage $= \sqrt{3} \times$ Phase voltage *i.e.,* $E_L = \sqrt{3} \cdot \times E_{ph}$

(ii) Line current = phase current *i.e.,* $I_L = I_{ph}$

TPS–7. DELTA OR MESH CONNECTIONS :

In this method of inter–connections, the dis–similar ends of three phase windings are connected together, *i.e.,* starting end of one phase is connected with finishing end of other phase and so on as shown in figure TPS–7.

The three line conductors run from the junctions as shown and outward directions are taken as positive.

The voltage across each winding or phase is called phase voltage and is represented by E_{ph} and voltage between any two lines is called line voltage and is represented by E_L.

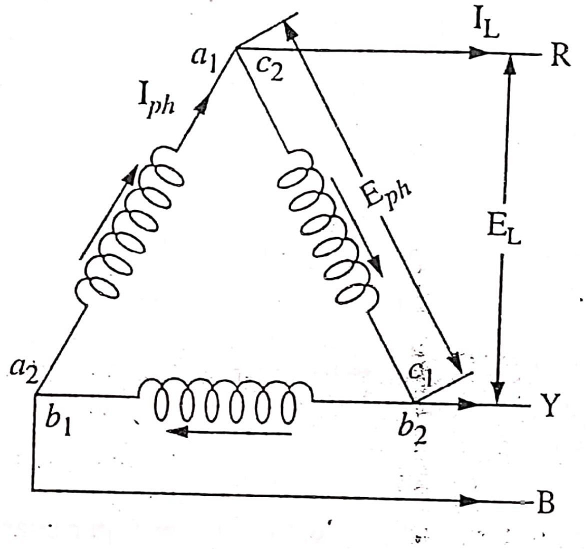


Fig. TPS–7

Similarly, current in each phase is called phase current and is written as I_{ph} and current in each as line current and is written as I_L.

TPS–8. RELATIONSHIP BETWEEN LINE VOLTAGE AND PHASE VOLTAGE IN A DELTA CONNECTED SYSTEM :

It is clear from the figure TPS–8, that in delta connections, the voltage between any pair of the lines is equal to the phase voltage of the phase winding connected between these two lines.

In other words, $E_L = E_{ph}$

Relationship between Line current and Phase current.

It is seen from the figure that

Current in line 1 is $(I_R - I_Y)$

Current in line 2 is $(I_B - I_R)$

Current in line 3 is $(I_Y - I_B)$

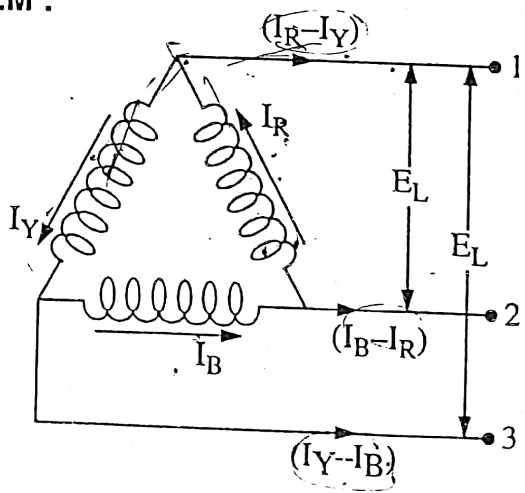


Fig. TPS–8

Current in each line is the difference of the two phase currents flowing through that line. Current in line 1 can be obtained by adding I_R and I_Y (reversed) *i.e.,* $I_R + (-I_Y)$. The angle between I_R and reversed I_Y is 60°

If $I_R = I_Y = I_B = I_{ph}$
and $(I_R - I_Y) = (I_B - I_R)$
$= (I_Y - I_B) = I_L$
then $(I_R - I_Y)^2 = I_R^2 + I_Y^2 + 2\, I_R .\, I_Y .\cos 60°$

or $I_L^2 = I_{ph}^2 + I_{ph}^2 + 2 I_{ph} . I_{ph} . \frac{1}{2}$

$\therefore \quad I_L = \sqrt{3} . I_{ph}$

All the line currents are equal in magnitude,

$\therefore \quad I_L = \sqrt{3} . I_{ph}$

From the vector diagram,

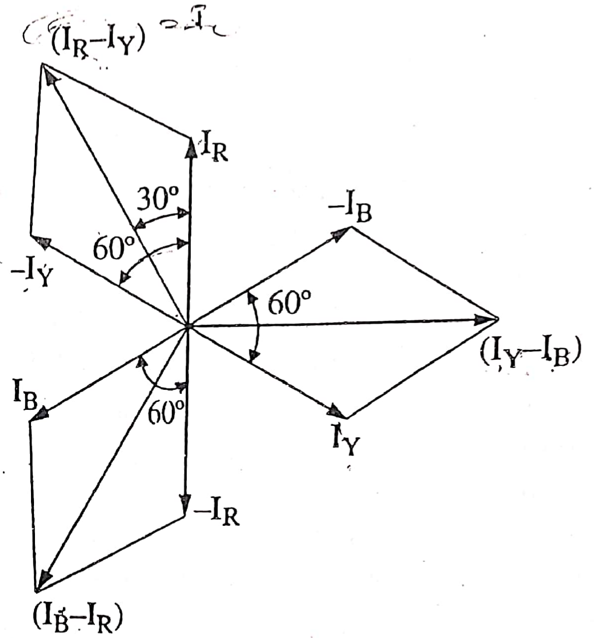


Fig. TPS–9

(i) the line currents are 120° apart.

(ii) line currents are 30° behind the corresponding phase currents.

Power. Total power in the delta connected system is the sum of the three phase powers *i.e.*,

Total power = 3 × phase power.

If E_{ph} is the *r.m.s.* value of *e.m.f.* per phase

I_{ph} is the *r.m.s.* value of current per phase

Power supplied per phase = $E_{ph} . I_{ph} . \cos \phi$

Total power supplied by 3–phases *i.e.* $P = 3 . E_{ph} . I_{ph} . \cos \phi$

$\therefore$ Line voltage, $E_L = E_{ph}$ and Line current $= \sqrt{3} . I_{ph}$

or $$I_{ph} = \frac{I_L}{\sqrt{3}}$$

$\therefore$ $$\text{Power} = 3 . E_L . \frac{I_L}{\sqrt{3}} . \cos \phi$$

or $$P = \sqrt{3} . E_L . I_L . \cos \phi \text{ watts.}$$

where ϕ is the phase angle between voltage and current.

Important relations in delta connections.

Line Voltage = phase voltage

i.e., $E_L = E_{ph}$

Line current = $\sqrt{3}$ × phase current

i.e., $I_L = \sqrt{3} \times I_{ph}$

... delta connected balanced system is

$$\left.\begin{array}{l} P = \sqrt{3} V_L . I_L . \cos\phi \\ \quad Or \\ P = \sqrt{3} E_L . I_L . \cos\phi \end{array}\right\} \text{line values} \qquad \left.\begin{array}{l} P = 3 V_{ph} . I_{ph} . \cos\phi \\ \quad Or \\ P = 3 E_{ph} . I_{ph} . \cos\phi \end{array}\right\} \text{phase values}$$

Problem TPS–1. 400 V (line to line) is applied to a three phase star connected load. Each side of the load has 4 ohms resistance in series with a 3 ohm reactance. Find *(i)* Line current *(ii)* Total Power consumed by the load.

Sol. Given,

Resistance, $R = 4\,\Omega$

Reactance, $X_L = 3\,\Omega$

Line voltage, $V_L = 400$ V

$\therefore$ Phase voltage, $V_{ph} = \dfrac{400}{\sqrt{3}} = 231\text{V}$

Impedence, $Z_{ph} = \sqrt{4^2 + 3^2} = 5\Omega$

Phase current, $I_{ph} = \dfrac{V_{ph}}{Z_{ph}} = \dfrac{231}{5} = 46{\cdot}2$ A.

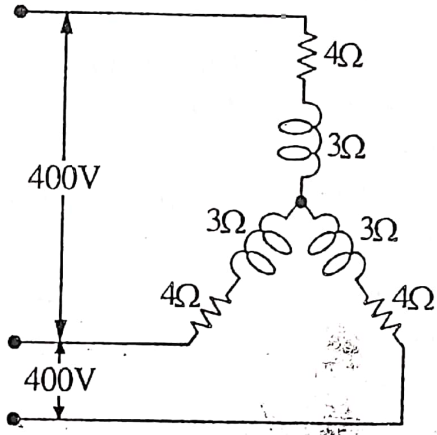


Fig. TPS–10

In star connections, Line current = Phase current.

$\therefore$ $I_L = I_{ph} = 46{\cdot}2$ amperes.

Power factor, $\cos\phi = \dfrac{4}{5} = 0\cdot 8$ (lagging)

Power consumed $= \sqrt{3} V_L . I_L . \cos\phi$

$= \sqrt{3} \times 400 \times 46\cdot 2 \times 0\cdot 8$

$= 25{,}600$ watts.

Problem TPS–2. Three similar resistors are connected in star across 440 volts, 3 phase lines. The line current is 10 amperes. Calculate the value of each resistor. To what value should the line voltage be changed to get the same line current with the resistors delta-connected ?

Sol. Given,

Line voltage, $V_L = 440$ V

Line current, $I_L = 10$ A

Star connections.

$I_L = I_{ph} = 10$ A

$V_{ph} = \dfrac{440}{\sqrt{3}} = 254\,\text{V}$

$R_{ph} = \dfrac{254}{10} = 25\cdot 4\Omega$

Delta connected.

$$I_L = 10 \text{ A}$$

$$I_{ph} = \frac{10}{\sqrt{3}} = 5 \cdot 77 \text{ A}$$

$$R_{ph} = 25{\cdot}4\ \Omega \text{ (already found)}$$

$$\therefore \quad V_{ph} = I_{ph} . R_{ph} = 5{\cdot}77 \times 25{\cdot}4 = \mathbf{146{\cdot}65\ V\ (Ans.)}$$

Note. *Voltage needed is* $\frac{1}{3}$ *rd, the voltage in star connections.*

Problem TPS–3. Three similar coils are star connected to a 3-phase, 400 V, 0 Hz supply. If inductance and resistance of each coil are 38·2 mH and 16 Ω espectively, determine,

(i) line current. (ii) power factor. (iii) power consumed.

Sol. Given,

Resistance, $R = 16\ \Omega$

Inductance, $L = 38{\cdot}2 \times 10^{-3}$ H

Line Voltage, $V_L = 400$ V

Frequency, $f = 50$ Hz

Inductive reactance, $X_L = 2\pi f . L$

$$= 2 \times 3{\cdot}142 \times 50 \times 38{\cdot}2 \times 10^{-3}$$

$$= 12\ \Omega$$

Impedence, $Z_{ph} = \sqrt{16^2 + 12^2} = 20\ \Omega$

In star connections.

$$\text{Phase voltage} = \frac{\text{Line voltage}}{\sqrt{3}}$$

$$V_{ph} = \frac{400}{\sqrt{3}} = 231 \text{V}$$

$$I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{231}{20} = 11 \cdot 55 \text{A}$$

$$I_L = I_{ph} = \mathbf{11{\cdot}55\ A\ \ Ans.}$$

Power factor, $\cos \phi = \frac{R}{Z} = \frac{16}{20} = \mathbf{0{\cdot}8\ \ Ans.}$

Power consumed, $P = \sqrt{3} V_L . I_L . \cos \phi$

$$= \sqrt{3} \times 400 \times 11 \cdot 55 \times 0 \cdot 8$$

$$= 6401 \text{ watts} = \mathbf{6{\cdot}401\ kW.\ \ Ans.}$$

Problem TPS–4. Three identical inductive loads of resistance 15 Ω and ctance 40 Ω are connected in delta across 440 V, 3-phase supply. Calculate,

(i) Ph...

Sol. Given,

Resistance, $R = 15\ \Omega$

Reactance, $X = 40\ \Omega$

Line voltage, $V_L = 440$ V

In delta connections.

$$V_{ph} = V_L = 440 \text{ V}$$

$$Z_{ph} = \sqrt{15^2 + 40^2} = 42\cdot7\Omega$$

(i) Phase current, $I_{ph} = \dfrac{V_{ph}}{Z_{ph}} = \dfrac{440}{42\cdot7} = 10\cdot3$ A (Ans.)

(ii) Line current, $I_L = \sqrt{3}\,.\,I_{ph} = \sqrt{3} \times 10\cdot3 = 17\cdot84$ A Ans.

(iii) Power absorbed, $P = \sqrt{3} V_L . I_L . \cos\phi$

$$= \sqrt{3} \times 440 \times 17\cdot84 \times \frac{15}{42\cdot7}$$

$$= 4776 \text{ watts.}$$

Problem TPS–5. Three similar impedences when connected in delta across a 3–phase, 400 V, 50 Hz system take a current of 13 A at a lagging power factor of 0·8 from the mains. Calculate the resistance and inductance of each impedence.

If the above three impedences are connected in star across the same supply as in above. Calculate *(i)* the current and *(ii)* the *k*VA taken from the supply.

Sol. Supply voltage, $V = 400$ volts.

Supply frequency, $f = 50$ Hz

Line current drawn, $I_L = 13$ amperes

Power factor, $\cos\phi = 0\cdot8$ (lagging)

Delta connections.

Phase current, $I_{ph} = \dfrac{I_L}{\sqrt{3}} = \dfrac{13}{\sqrt{3}} = 7\cdot5$ amperes.

Phase voltage, $E_{ph} = E_L = 400$ volts

Phase impedence, $Z_{ph} = \dfrac{E_{ph}}{I_{ph}} = \dfrac{400}{7\cdot5} = 53\cdot3$ ohms

Since power factor, $\cos\phi = \dfrac{R_{ph}}{Z_{ph}}$

$\therefore$ Resistance/phase, $R_{ph} = Z_{ph} . \cos\phi$

$= 53\cdot3 \times 0\cdot8 = 42\cdot6$ ohms (Ans.)

Reactance/phase, $X_{ph} = Z_{ph} \sin\phi$

$= 53\cdot3 \times 0\cdot6 = 32$ ohms Ans.

Also $X_{ph} = 2\pi f . L$

or $\quad 32 = 2 \times \pi \times 50 \times L$

$\therefore \quad L = \dfrac{32}{2 \times \pi \times 50} = 0\cdot106$ Henry. **Ans.**

Each impedence has a resistance of 42·6 ohms and an inductance of 0·106 Henry.

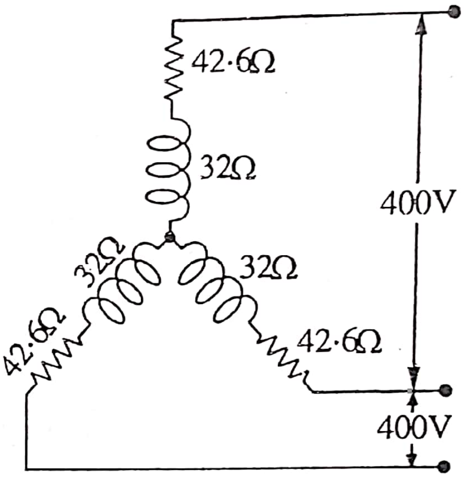


Fig. TPS–11

Star connections.

Line voltage, $\quad E_L = 400$ volts

Phase impedence, $\quad Z_{ph} = 53\cdot3$ ohms

Phase voltage, $E_{ph} = \dfrac{400}{\sqrt{3}} = 231$ volts

Phase current, $I_{ph} = \dfrac{E_{ph}}{Z_{ph}} = \dfrac{231}{53\cdot3}$

$= 4\cdot33$ amperes.

$\therefore \quad I_L = I_{ph} = 4\cdot33$ amperes **Ans.**

Power Input in $kVA = \sqrt{3} \times E_L . I_L = \sqrt{3} \times 400 \times 4 \cdot 33$

$= 3000$ volt–amperes $= 3$ kVA **Ans.**

Problem TPS–6. Three inductive coils of 0·05 H are connected in star to a 3–phase 200 volts, 50 Hz system. Calculate the inductance of 3 coils which when connected in delta to the same system, will take the same line current.

Sol. Inductance of each coil, $\quad L = 0\cdot05$ H

Taking voltage as phase voltage, $E_{ph} = 200$ V

Supply frequency, $\quad f = 50$ Hz

Line voltage, $\quad E_L = 200 \times \sqrt{3}$ V

Inductive reactance of the coil, $X_L = 2\pi f.\ L$

$= 2 \times \pi \times 50 \times 0\cdot05$

$= 15\cdot7$ ohm

$\therefore \quad$ Phase current, $I_{ph} = \dfrac{200}{15\cdot7} = 12\cdot74$ amperes

Now in star connections, line current is equal to the phase current.

$\therefore \quad I_L = I_{ph} = 12\cdot74$ amperes.

In the problem, the line current required in delta system is the same as the line current of star *i.e.*, 12·74 amperes.

Now the phase current, $I_{ph} = \dfrac{I_L}{\sqrt{3}}$ in delta connections.

$I_{ph} = \dfrac{12 \cdot 74}{\sqrt{3}}$ amperes.

i.e., $E_{ph} = E_L = 200 \times \sqrt{3}$

Inductive reactance per phase in delta system,

$$X_L = \frac{E_{ph}}{I_{ph}} = \frac{200 \times \sqrt{3}}{\frac{12 \cdot 74}{\sqrt{3}}} = 47 \cdot 1 \text{ ohms.}$$

$\because$ $X_L = 2\pi f. L$

$$\therefore \quad L = \frac{X_L}{2\pi f} = \frac{47 \cdot 1}{2 \times \pi \times 50} = 0{\cdot}1488 \text{ Henry (Ans.)}$$

Problem TPS–7. A 100 H.P. (metric), there phase, star connected motor works on a supply whose line voltage is 3000 volts. The motor efficiency is 92% and power factor is 0·9 (lagging). Calculate the line current.

Sol. Output of the motor, $P = 100$ H.P.

$= 100 \times 735{\cdot}5 = 73550$ watts

Efficiency of the motor, $\eta = 92\%$

Power factor of the motor, $\cos\phi = 0{\cdot}9$

Motor Input Power, $W = \dfrac{\text{output}}{\text{efficiency}}$

$= \dfrac{73550}{0 \cdot 92} = 79{,}945 \cdot 6$ watts

Since input power, $W = \sqrt{3}\,.\,E_L\,.\,I_L\,.\cos\phi$

$\therefore$ $79{,}945{\cdot}6 = \sqrt{3} \times 3000 \times I_L \times 0 \cdot 9$

$\therefore$ $I_L = 17$ amperes. **Ans.**

Problem TPC-8. A 3-phase supply system with 250 V between each phase and neutral supplies a balanced 3-phase star connected load having a resistance of 10 Ω in each phase. Calculate the current flowing in each phase and the neutral. Also find the power consumed.

Sol. Phase voltage, $V_{ph} = 250$ V

Load Resistance, $R_{ph} = 10\ \Omega$

Since all the three resistances of 10 ohm each are connected in star as shown in the fig.

$\therefore$ Current in each phase, $I_{ph} = \dfrac{V_{ph}}{R_{ph}}$

$$\therefore \quad I_{ph} = \frac{250}{10} = 25\text{A}$$

As the 3-phase load is a balanced load. Therefore current flowing through each phase is the same and are displaced by 120° electrical as shown in fig.

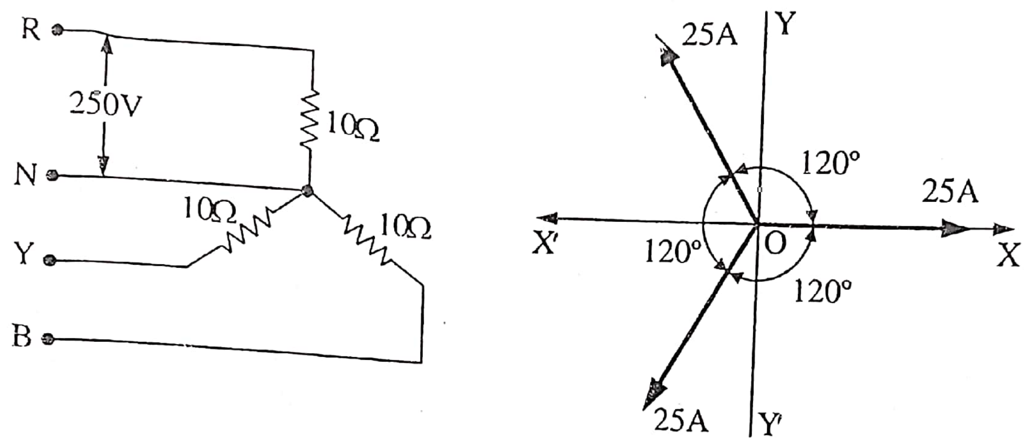


Fig. TPS—12

The current flowing through the neutral will be the vector sum of all the three ıse currents.

Resolving the phase currents horizontally and vertically,

$$
\begin{aligned}
I_{xx} &= 25 \cos 0° - 25 \cos 60° - 25 \cos 60° \\
&= 25 - 25 \cos 60° - 25 \cos 60° \\
&= 25\,(1 - 2 \cos 60) \\
&= 25\,(1 - 2 \times 1/2) = 0
\end{aligned}
$$

and

$$
\begin{aligned}
I_{yy} &= 25 \sin 0 + 25 \sin 60 - 25 \sin 60° \\
&= 0 + 25 \sin 60° - 25 \sin 60° = 0
\end{aligned}
$$

$\therefore$ Current in Neutral, $I_N = \sqrt{(I_{xx})^2 + (I_{yy})^2}$

$= \sqrt{0 + 0}$ = **zero Ans.**

Therefore power absorbed,

$P = 3\, V_{ph}\, I_{ph} \cos \phi$ $\quad (\because \cos \phi = 1)$

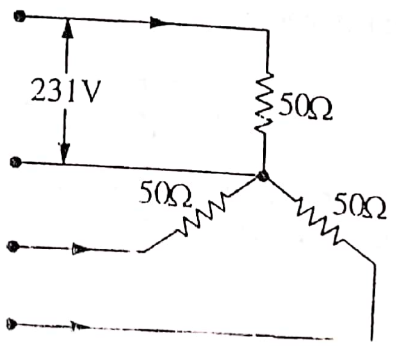
$\therefore$ $P = 3 \times 250 \times 25 \times 1$ 18750 watts

$= 18{\cdot}75\ kW$ **Ans.**

Problem TPS-9. Three 50 ohms non inductive resistors are connected in (i) (ii) delta, across a 400 V, 50 c/s supply. Calculate the power taken from the ›ly in each case. In the event of one of the resistance getting opened, what ld be the value of the total power taken from the supply ?

Sol. Resistance of each resistor, $R = 50\ \Omega$

Line voltage or supply voltage, $V_L = 400$ V



Case No. I

When all the three resistors are connected ar as shown in fig. TPS-12(a)

Voltage across each resistor, $V_{ph} = \dfrac{400}{\sqrt{3}} = 231$ V.

Phase current I_{ph} $\dfrac{231}{\ }$ = 4·62 A

Power taken from the supply, $P = 3\ V_{ph} \cdot I_{ph} \cdot \cos\phi$ $\left[\begin{array}{l} \because \text{load is pure resistance} \\ \therefore \cos\phi = 1 \end{array}\right]$

$$= 3 \times 231 \times 4{\cdot}62 \times 1$$

$\therefore$ **P = 3201·66 watts Ans.**

When one of the resistors is disconnected.

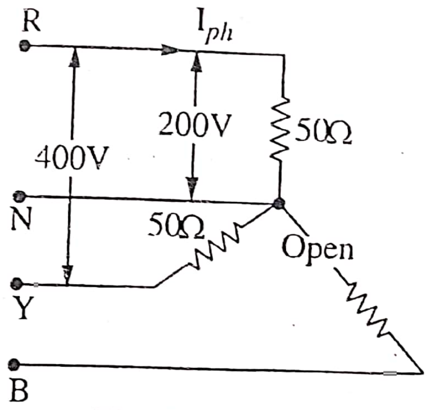


Fig. TPS-12(b)

Two Resistors are connected in series across terminals R and Y

Voltage across terminals R and Y is,

$$V_L = 400\ V$$

Total Resistance between R and Y is,

$$R_T = 50 + 50 = 100\ \Omega$$

$\therefore$ Current through each resistor $= \dfrac{400}{100} = 4A$

Power taken from supply, $P = 2\ V_{ph} \cdot I_{ph} \cdot \cos\phi$

$\therefore$ $P = 2 \times 200 \times 4 \times 1 =$ **1600 watts (Ans.)**

Case No. II. When all the three resistances are connected in delta as shown in fig. TPS-12(c)

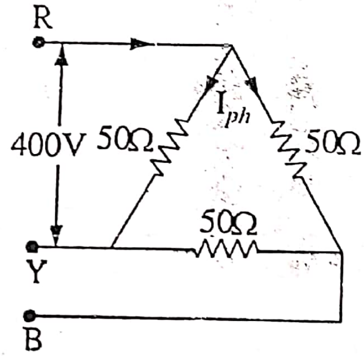


Fig. TPS-12(c)

Voltage across each resistor, $V_{ph} = V_2 = 400\ V$

Resistance of each phase, $R_{ph} = 50\ \Omega$

Current in each phase, $I_{ph} = \dfrac{400}{50} = 8A$

Power taken from supply, $P = 3\ V_{ph} \cdot I_{ph} \cdot \cos\phi$

$$= 3 \times 400 \times 8 \times 1$$

$$= \textbf{9600 watts Ans.}$$

When one of the resistors is disconnected as shown in the fig. TPS-12(d)

Here, only two resistors are in the circuit, the voltage across each is same.

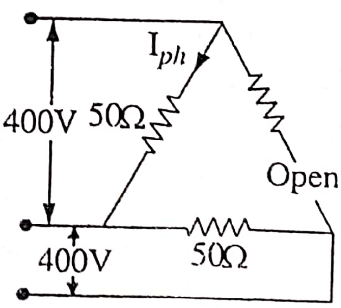


Fig. TPS-12(d)

Voltage across each resistor, $V_{ph} = V_L = 400\ V$

Resistance per phase, $R_{ph} = 50\ \Omega$

Current per phase, $I_{ph} = \dfrac{400}{50} = 8A$

Power taken from supply, $P = 2\ V_{ph} \cdot I_{ph} \cdot \cos\phi$

$\therefore$ Power, $P = 2 \times 400 \times 8 \times 1$

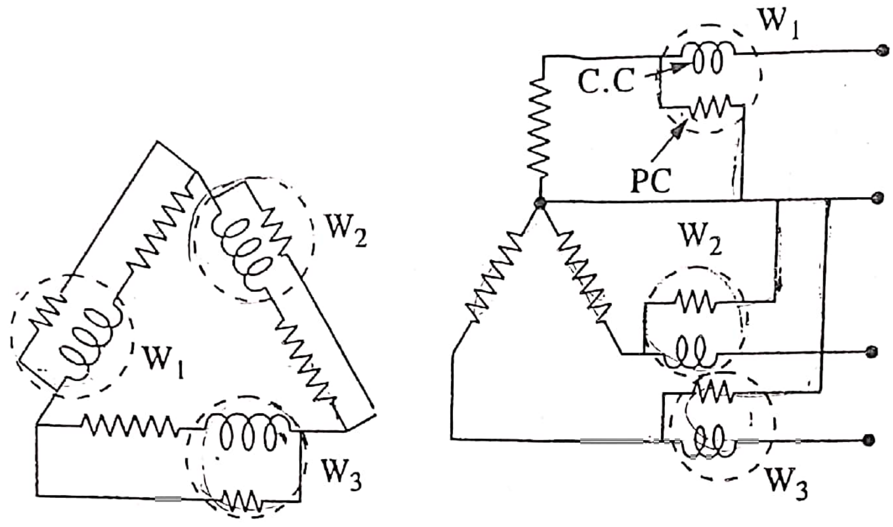
$$= \textbf{6400 watts Ans.}$$

TPS–9. MEASUREMENT OF POWER IN THREE PHASE CIRCUITS :

The various methods which are used for the measurement of power in 3-phase circuits are as under :-

(i) **Three Wattmeter Method.** In this method, one watt meter is connected in

each phase as shown in the figure TPS-13.



Delta Connected **Star Connected**

Fig. TPS–13

The total power is the algebraic sum of the readings of all the three watt meters. This method is used in special cases. If the alternator is star connected and the star point or neutral is not available, an artificial star point can be made by connecting three high resistances in star to the three line conductors.

The current coil (CC.) of the watt meter is connected in series with the line carrying the current and the pressure coil (P.C.) is connected across the two points whose potential difference is to be measured as shown in figure TPS–13 *(a)*. Thus the watt meter gives the reading which is the product of current through its circuit coil, the *p.d.* across its pressure coil and the cosine of the angle between this voltage and current.

In this method, three watt meters, W_1, W_2 and W_3 are connected in each of the three phases of the load as shown in figure TPS–13 *(a)*. Whether delta connected or star connected the current coil of each watt meter carries the current of one phase only and pressure coil measures the phase voltage of this phase. So in this way; each watt meter reads or measures the power of each phase.

Hence Total Power, $P = W_1 + W_2 + W_3$

Two Wattmeter Method. This method gives the actual power or real power in 3-phase circuit, even if the neutral of the load is not connected to the neutral of the source. However, it is possible only if the load is perfectly balanced. This method can also be used for 3-phase, 4 wire system in which the neutral wire carries the neutral current.

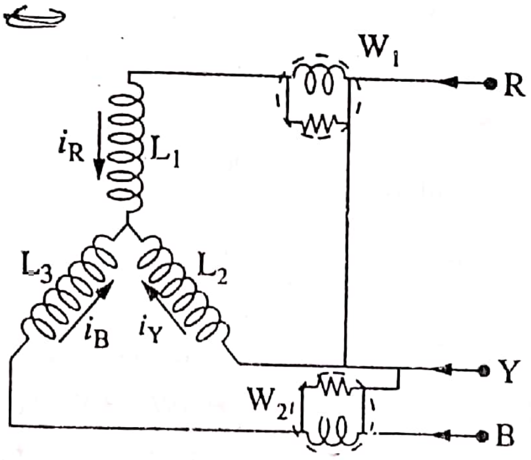


Fig. TPS–14

. **Two Wattmeter Method. (Balanced or unbalanced load)** ; In this method, two Watt meters are used to measure the power in 3 phase circuit. The figure TPS–14 shows the connection diagram of such a circuit.

Let the instantaneous voltage measured from the star point to the three line wires be e_R, e_Y and e_B respectively and let the instantaneous current flowing in the three coils be i_R, i_Y and i_B respectively. The current passing through the current coil of W_1 is i_R whereas the voltage across the pressure coil of W_1 is $(e_R - e_Y)$.

Similarly the current flowing through the current coil of W_2 is i_B and the voltage across the pressure coil is $(e_B - e_Y)$.

$$\therefore \quad W_1 + W_2 = (e_R - e_Y).\, i_R + (e_B - e_Y).\, i_B$$

$$= e_R.\, i_R + e_B.\, i_B - e_Y\,(i_R + i_B)$$

Now $\quad i_R + i_Y + i_B = 0 \qquad$ (Kirchhof's first law)

$$\therefore \quad i_Y = -(i_R + i_B)$$

$$\therefore \quad W_1 + W_2 = e_R\, i_R + e_Y.\, i_Y + e_B.\, i_B$$

$$= p_1 + p_2 + p_3$$

Where p_1 is the power absorbed by load L_1, p_2 that absorbed by L_2 and p_3 that absorbed by L_3.

This proof is true whether the load is balanced or unbalanced. If the load is star connected, then it should not have neutral connection (3 – ϕ, 3 wire system) and if it has a neutral wire (3ϕ, 4 wire system) then it should be exactly balanced. So that in each case, there is no neutral current I_N other wise Kirchhof's first law will give,

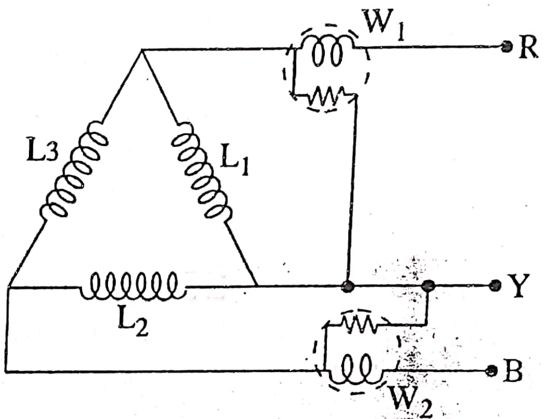


Fig. TPS–15

$$i_N + i_R + i_Y + i_B = 0$$

The ~~same~~ result holds good for a delta connection.

Imp

~~Two~~ Watt meter Method. (Balanced load) ; Let E_R, E_Y and E_B be the R.M.S. values of the three phase voltages and I_R, I_Y and I_B be the R.M.S. values of load currents. Each watt meter has its current coil in series with one of the lines and the pressure coils have one of their ends connected to one end of the corresponding current coil, while the other is connected to the third line which has no current coil as shown in figure TPS–16 *(a)*. These voltages and currents are assumed sinusoidal. Therefore, they can be represented by vectors. The load may be assumed inductive and so the currents would lag behind their phase voltages.

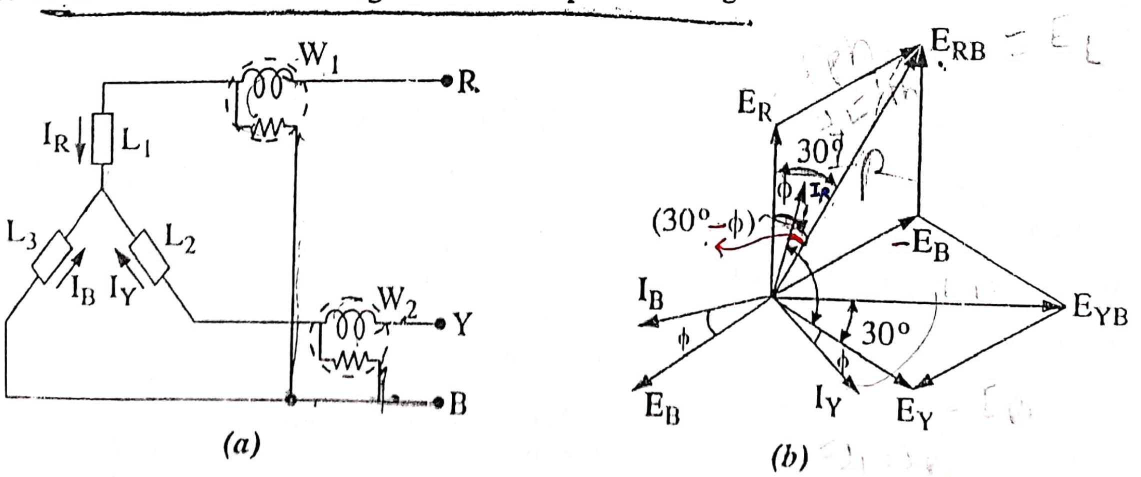


Fig. TPS–16

The vector diagram for a balanced star connected load is as shown in figure TPS-16 (*b*) in which the voltages and currents are taken in terms of R.M.S. values.

Current through watt meter W_1 is I_R.

$\therefore$ P.D. across pressure coil of $W_1 = (E_R - E_B)$

or $E_{RB} = (E_R - E_B)$.... vectorially

By adding E_R and E_B (reversed) as shown in figure TPS-16 (*b*). The phase difference between E_{RB} and $I_R = (30 - \phi)$

Similarly current through W_2 is I_Y.

P.D. across the pressure coil of W_2 is, $E_{YB} = (E_Y - E_B)$ ...Vectorially

Again E_{YB} is obtained by compounding E_Y and E_B (reversed) as shown in figure TPS-16 (*b*). The phase difference between E_{YB} and $I_Y = (30 + \phi)$.

Now $E_{RB} = E_{YB} = E_L$ (line voltage)

and $I_R = I_Y = I_B = I_L$ ($\because$ star connections)

$\therefore$ $W_1 = E_L \cdot I_L \cdot \cos(30 - \phi)$

and $W_2 = E_L \cdot I_L \cdot \cos(30 + \phi)$

Total Power, $W_1 + W_2 = E_L \cdot I_L \cos(30 - \phi) + E_L \cdot I_L \cos(30 + \phi)$

$$\therefore \quad W_1 + W_2 = E_L \cdot I_L \{\cos(30 - \phi) + \cos(30 + \phi)\}$$

$$= E_L \cdot I_L \cdot \{\cos 30 \cos \phi + \sin 30 \sin \phi\} + \{\cos 30 \cos \phi - \sin 30 \sin \phi\}$$

$$= E_L \cdot I_L \{2 \cos 30 \cos \phi\}$$

$$= \sqrt{3} E_L \cdot I_L \cos \phi = \text{Total Power}$$

Power Factor From Watt-meter Readings

It has been proved above that for a lagging power factor,

$$W_1 = E_L \cdot I_L \cdot \cos(30 - \phi)$$

and

$$W_2 = E_L \cdot I_L \cdot \cos(30 + \phi)$$

$$\therefore \quad W_1 + W_2 = E_L \cdot I_L \cdot \cos(30 - \phi) + E_L \cdot I_L \cos(30 + \phi)$$

$$= E_L \cdot I_L \{\cos 30 \cos \phi + \sin 30 \sin \phi + \cos 30 \cos \phi - \sin 30 \sin \phi\}$$

i.e. $$W_1 + W_2 = \sqrt{3} E_L \cdot I_L \cos \phi \qquad \text{...(i)}$$

Similarly, $W_1 - W_2 = E_L \cdot I_L \cdot \cos(30 - \phi) - E_L \cdot I_L \cdot \cos(30 + \phi)$

or $$W_1 - W_2 = E_L \cdot I_L \cdot \sin \phi \qquad \text{... (ii)}$$

Dividing (*ii*) by (*i*), we get,

$$\frac{W_1 - W_2}{W_1 + W_2} = \frac{E_L \cdot I_L \cdot \sin \phi}{\sqrt{3} E_L \cdot I_L \cdot \cos \phi}$$

or $$\tan \phi = \frac{\sqrt{3} \cdot (W_1 - W_2)}{W_1 + W_2}$$

From this, it is clear that individual readings of the watt meters not only depend

We will consider the following cases :

(a) When $\phi = 0$ *i.e.*, power factor = 1 (Resistive load).

Then, $W_1 = W_2 = E_L . I_L . \cos 30°$

Hence the readings of both the watt meters will be same.

(b) When $\phi = 60°$ *i.e.*, power factor = 0·5

$W_1 = E_L . I_L . \cos (30 - 60)$ and

$W_2 = E_L . I_L . \cos (30 + 60) = 0$

Thus the power will be measured by W_1 only.

(c) When $\phi > 60°$ *i.e.*, power factor is < 0·5, then

W_1 is positive but W_2 will be negative.

(d) When ϕ

$= 90°$ *i.e.*, power factor is 0.

Then, $W_1 = E_L . I_L . \cos (30 - 90)$

and $W_2 = E_L . I_L . \cos (30 + 90)$

The two readings are equal but of opposite sign.

$\therefore \quad W_1 + W_2 = 0$

significan

One Watt meter Method. This method is used only for balanced loads. The connections for a star system are shown in figure TPS-17. The ~~current coil~~ of watt meter is connected in series with one of the line and the pressure coil is connected alternately between this line and the other two lines. ~~across~~ the two lines. The two readings obtained correspond to those obtained by normal two wattmeter method.

This method is sometimes used to check the load on the motors.

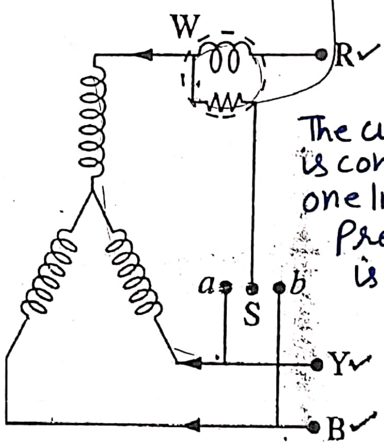


Fig. TPS–17

The current coil is connected in any one line and the Pressure coil is connected alternately between this and the other two lines.

Problem TPS–10. Two wattmeters connected to measure the input to a balanced 3-phase circuit indicate 2000 W and 500 W respectively. Find the power factor of the circuit *(i)* when both readings are positive *(ii)* when latter reading is obtained after reversing the connection to the current coil of one instrument.

Sol. Reading of first watt meter, $W_1 = 2000$ watts

Reading of second watt meter, $W_2 = 500$ watts

(i) Total power, $W_1 + W_2 = 2000 + 500 = 2500$ watts

Also $W_1 - W_2 = 2000 - 500 = 1500$ watts

$$\therefore \quad \tan\phi = \sqrt{3}\left(\frac{W_1 - W_2}{W_1 + W_2}\right) = \sqrt{3}\left(\frac{1500}{2500}\right) = 1 \cdot 0392$$

$$\therefore \quad \phi = \tan^{-1} 1 \cdot 0392 = 46° \, 6'$$

Hence $\cos\phi = \cos 46° \, 6' = \mathbf{0 \cdot 6394}$ **(Ans.)**

(ii) When the connections of second watt meter are reversed, *i.e.*, considering W_2 as negative.

$$\therefore \quad \tan\phi = \sqrt{3}\,\frac{W_1 + W_2}{W_1 - W_2} = \sqrt{3}\,\frac{2500}{1500} = 2\cdot 887$$

$$\therefore \quad \phi = \tan^{-1} 2\cdot 887 = 70°\ 54'$$

Hence $\cos\phi = \cos 70°\ 54' = \mathbf{0\cdot 327}$ **(Ans.)**

Problem TPS—11. The power input to a 500 H.P. 1100 V, 3 – ϕ induction motor running at full load is measured by two wattmeters which indicate 290 watts and 125 watts respectively. Calculate,

(i) **Power Input No. 1** ***(ii)*** **Power factor**

(iii) **Line current** ***(iv)*** **Efficiency.**

Sol. Motor output $= 500$ H.P. $= 500 \times 0\cdot 735 = 367\cdot 5\ kW$

Reading of watt meter No-1, $W_1 = 125$ watts

Reading of second watt meter No-2, $W_2 = 290$ watts

Supply voltage, $V_L = 1100$ volts

(i) Power Input to the motor, $P = W_1 + W_2$

$= 125 + 290 = \mathbf{415\ kW\ Ans.}$

(ii)
$$\tan\phi = \sqrt{3}\left(\frac{W_2 - W_1}{W_2 + W_1}\right) = \sqrt{3}\left(\frac{290 - 125}{290 + 125}\right)$$

or
$$\tan\phi = \sqrt{3} \times \frac{165}{415} = 0\cdot 6886$$

$$\therefore \quad \phi = \tan^{-1} 0\cdot 6886 = 34°33'$$

Power factor, $\cos\phi = \cos 34°33' = 0\cdot 82$ **Ans.**

(iii) Line current,
$$I_L = \frac{\text{Power Input in watts}}{\sqrt{3} \times \text{line voltage} \times p.f.}$$

$$\therefore \quad I_L = \frac{415 \times 1000}{\sqrt{3} \times 1100 \times 0\cdot 82} = 266 \text{ Amp.}$$

(iv)
$$\%\ \text{Efficiency} = \frac{\text{Motor output}}{\text{Motor Input}} \times 100 = \frac{367\cdot 5}{415} \times 100 = \mathbf{88\cdot 55\%\ Ans.}$$

Problem TPS—12. The power input to a synchronous motor is measured by two watt meters, both of which reads 50 kW. If the power factor of the motor be changed to 0·866 leading, determine the reading of the two. Draw the vector diagram for the second condition of the load.

Sol. The reading of the two watt meters are given as,

$$W_1 = V_L \cdot I_L \cos(30 + \phi)$$
$$W_2 = V_L \cdot I_L \cos(30 - \phi)$$

In the first case both watt meters read equal and positive,

when $\cos(30 + \phi) = \cos(30 - \phi)$

or $\cos 30 \cos\phi - \sin 30 \sin\phi = \cos 30 \cos\phi + \sin 30 \sin\phi$

or $2 \sin 30 \sin\phi = 0$ or $\sin\phi = 0$

$\therefore \qquad \phi = \sin^{-1} 0 = 0$

or Power factor of the load, $\cos \phi = \cos 0 = \mathbf{1}$ **Ans.**

Case No. II. When the power factor be changed to 0·866 leading.

Here, $\cos \phi = 0{\cdot}866 \quad \therefore \quad \phi = \cos^{-1} 0{\cdot}866 = 30°$

$$\tan 30° = \frac{1}{\sqrt{3}}$$

Also $$\tan \phi = \sqrt{3}\left(\frac{W_2 - W_1}{W_2 + W_1}\right) \qquad \text{But } W_2 + W_1 = 100 \qquad \ldots(i)$$

$$\therefore \quad \frac{1}{\sqrt{3}} = \sqrt{3}\left(\frac{W_2 - W_1}{W_2 + W_1}\right) \quad \text{or} \quad \frac{1}{3} = \frac{W_2 - W_1}{50+50}$$

$$\text{or} \quad \frac{1}{3} = \frac{W_2 - W_1}{100} \quad \text{or} \quad W_2 - W_1 = \frac{100}{3} \qquad \ldots(ii)$$

Solving *(i)* and *(ii)* we get,

$$W_2 + W_1 = 100 \qquad \text{Also} \quad W_2 - W_1 = \frac{100}{3}$$

$$\therefore \quad 2W_2 = \frac{400}{3} \quad \therefore \quad W_2 = \frac{400}{6} = \mathbf{66{\cdot}67}\ \boldsymbol{k}\ \textbf{watts Ans.}$$

Since, $W_2 + W_1 = 100 \quad \therefore \quad W_1 = 100 - 66{\cdot}67 = \mathbf{33{\cdot}33}\ \boldsymbol{k}$ **watts Ans.**

Problem TPS—13. A star connected balanced load is supplied from a 3-phase supply with a line voltage of 440 volts at a frequency of 50 c/s. Each phase of load consists of a resistance and a capacitor joined in series and the readings on two wattmeters connected to measure the total power supplied are 782 watts and 1980 watts, both positive. Calculate (*a*) the power factor of the circuit (*b*) the line current (*c*) the capacitance of each capacitor.

Sol. Supply voltage or line voltage, $V_L = 440$ V

Reading of First watt meter, $W_1 = 782$ watts

Reading of second watt meter, $W_2 = 1980$ watts

From the formula, $$\tan \phi = \sqrt{3}\left(\frac{W_2 - W_1}{W_2 + W_1}\right)$$

$$\text{or} \quad \tan \phi = \sqrt{3}\left(\frac{1980 - 782}{1980 + 782}\right) = \sqrt{3} \times \frac{1198}{2762}$$

or $\tan \phi = 0{\cdot}75$

$\therefore \quad \phi = \tan 0{\cdot}75 = 36° \, 54'$

(*a*) Power factor of the circuit, $\cos \phi = \cos 36° \, 54' = \mathbf{0{\cdot}8}$ **(lead)**

(*b*) Power in 3-phase circuit, $P = W_1 + W_2 = \sqrt{3}.V_L.I_L.\cos\phi.$

$$\therefore \quad 782 + 1980 = \sqrt{3} \times 440 \times I_L \times 0{\cdot}8$$

In star connections, $I_L = I_{ph} = 4{\cdot}53$ A

(c) Now phase voltage, $V_{ph} = \dfrac{V_L}{\sqrt{3}}$

$$\therefore \quad V_{ph} = \frac{440}{\sqrt{3}} = 254\,\text{V} \qquad \therefore \quad Z_{ph} = \frac{V_{ph}}{I_{ph}} = \frac{254}{4\cdot53} = 56 \text{ ohms}$$

and $\quad R_{ph} = Z_{ph} \cos\phi = 56 \times 0{\cdot}8 = 44{\cdot}56$

$$\therefore \quad X_c \,|_{\text{phase}} = \sqrt{Z_{ph}^2 - R_{ph}^2} = \sqrt{(56)^2 - (44\cdot86)^2} = 33\cdot52\Omega$$

$\therefore$ Capacitive Reactance/phase, $X_c = 33{\cdot}52$ ohms

Also $$33{\cdot}52 = \frac{1}{2\pi \times 50 \times C}$$

$$\therefore \quad C = \frac{10^6}{2 \times 3\cdot14 \times 50 \times 33\cdot52} = 95\ \mu\text{F (Ans.)}$$

Problem TPS—14. A 50 V, 3–phase motor has an output of 37·3 kW and operates at a power factor of 0·85 lagging with an efficiency of 90%. Calculate the reading on each watt meter connected to measure the input.

Sol. Motor output $= 37{\cdot}3 \times 1000 = 37300$ W

Supply voltage, $V = 500$ Volts

Motor efficiency, $\eta = 90\%$

Power factor, $\cos\phi = 0{\cdot}85$ (lag)

$$\therefore \quad \text{Motor Input} = \frac{\text{Motor output}}{\text{efficiency}} = \frac{37300}{0\cdot9} = 41{,}400 \text{ watts}$$

$\therefore$ Total Power Input, $W_1 + W_2 = 41{,}400$ watts ...(*i*)

Now $\quad \cos\phi = 0{\cdot}85$

$\therefore \quad \phi = \cos^{-1} 0{\cdot}85 = 31°\ 48'$

$\therefore \quad \tan\phi = \tan 31°\ 48' = 0{\cdot}62$

or $$0\cdot62 = \sqrt{3}\,\frac{(W_1 - W_2)}{W_1 - W_2} = \frac{\sqrt{3}\,(W_1 - W_2)}{41400}$$

$\therefore \quad W_1 - W_2 = 14860$ ...(*ii*)

From (*i*) and (*ii*), we get,

$$W_1 = 28\cdot15\ kW, \quad W_2 = 13\cdot29\ kW.$$

Problem TPS–15. If the readings on two wattmeters, connected to a 3–phase balanced load are 5 kW and 0·5 kW, the latter being obtained after reversal of the current coil connections. Calculate the power and power factor of the load.

Sol. Reading of first Watt meter, $W_1 = 5\ kW$

Reading of second Watt meter, $W_2 = 0{\cdot}5\ kW$

$$W_1 + W_2 = 5 + 0.5 = 5.5\ kW \quad \textbf{Ans.}$$

Since the connection of the latter are reversed, so take the reading of W_2 as negative.

$$\therefore \quad \tan\phi = \sqrt{3}.\left(\frac{W_1 + W_2}{W_1 - W_2}\right) = \sqrt{3} \times \frac{5 \cdot 5}{4 \cdot 5} = 2 \cdot 117$$

$$\therefore \quad \phi = \tan^{-1} 2{\cdot}117 = 64°\ 43'$$

Hence $\cos\phi = \cos 64°\ 43' = 0{\cdot}42$ **Ans.**

$\therefore$ The Power factor of the load is 0·42

SUMMARY

1. Polyphase system. It is an A.C system in which a group of two or more than two equal voltages of same frequency are arranged in such a way that these three voltages have same phase difference between them. This system is known as polyphase system.

2. Advantages of three phase system over single phase system are ; *(a)* Constant power for balanced load *(b)* higher rating *(c)* more efficiency of 3 phase motor *(d)* more efficient *(d)* economy in power transmission.

3. Phase sequence. The order in which the voltages (*e.m.f's*) reach their maximum values.

4. Generation of 3 phase *e.m.f's*

$$e_{a_1a_2} = E_{max} \sin \omega t$$

$$e_{b_1b_2} = E_{max} \sin (\omega t - 120°)$$

$$e_{c_1c_2} = E_{max} \sin (\omega t - 240°)$$

5. Interconnections of three phases.

(a) Star connection, $E_L = \sqrt{3}E_{ph}, \quad I_L = I_{ph}$

(b) Delta connection, $E_L = E_{ph}, \quad I_L = \sqrt{3}\, I_{ph}.$

(c) Power in three phase circuit $= 3\, V_{ph} I_{ph}. \cos\phi = \sqrt{3}\, V_L . I_L \cos\phi$

6. Power measurement in 3-phase circuit.

(a) By three wattmeter method

(b) By two wattmeter method

$$W_1 = V_L.I_L.\cos(30 - \phi)$$

$$W_2 = V_L.\ I_L.\cos(30 + \phi)$$

$$\text{Power} = W_1 + W_2 = \sqrt{3}\ V_L.\ I_L \cos\phi$$

$$\tan\phi = \sqrt{3}\left(\frac{W_1 - W_2}{W_1 + W_2}\right)$$

(c) By one watt meter method

SHORT ANSWER QUESTIONS

Q. 1. What do you understand by three phase balanced load ?

Ans. The three phase loads will be balanced when the impedance connected in three phases are same in magnitude as well as in phase.

Q. 2. What is phase sequence in a 3- phase system ?

Ans. Phase sequence is an order or sequence in which voltage in three different phases attain their maximum values one after the other.

Q. 3. Why interconnection of 3-phase system is necessary ?

Ans. In non-interconnected three phase system, we require 6 wires ; while in inter connected 3- phase system we require 3 or 4 wires. Therefore, non-inter connected system will be complicated and expensive also.

Q. 4. How the given sequence is reversed ?

And. The given phase sequence can be reversed by inter changing any two terminals of the supply.

Q. 5. Why delta connected system cannot be used for supplying lighting as well as power loads.

Ans. Since in delta connected system there is no neutral, therefore, we cannot provide 3 phase, 4 wire system. So due to the absence of neutral, we can not supply lighting as well as power loads because neutral is essential for supplying these loads.

Q. 6. Name the electrical equipments in each case having unity power factor and lagging power factor.

Ans. The electrical equipments operating at unity power factor are ;

(i) Incandescent lamps and electric heaters

The electrical equipments operating at lagging power factor are Induction furnaces, arc lamps etc.

Q. 7. While measuring power in a balanced three phase circuit, one of the wattmeters indicates zero, as the load is varied. How do you account for it ?

Ans. One of the watt meters indicate zero as the load is varied if and only if the power factor of the load remains constant at 0·5. The reading of the other wattmeter will vary with the variation in load.

Q. 8. Differentiate between star and Delta connections

Ans. In star connections, similar terminals (*either start or finish*) of the three phases are connected together to provide star or neutral point, whereas in delta connected system, start end of the one phase is connected to finish terminal of the second phase and start terminal of second phase is connected to the finish terminal of the third phase and so on to provide closed circuit in 3-phase system.

Q. 9. **Why is an unbalanced load not normally used on a 3- phase-3 wire system ? Are any line/phase voltages equal in such situations ?**

Ans. Normally unbalanced load is not used on 3- phase-3 wire ungrounded star connected system because unbalanced loading can cause different voltages drop in the lines. Therefore the three phase voltages can be different in magnitude as well as in phase and it is also possible that one phase voltage may exceed the line voltage. This is not desirable. Due to different voltage levels, some loads may operate inefficiently due to lowering of the voltage and the other equipment may get damaged due to over voltage.

OBJECTIVE QUESTION BANK

Type–I Fill in the Blanks :

1. In a 3–phase system, the phase difference between the two adjacent *e.m.f.*'s ....
2. With star connections
   (1) I_L = ... (2) V_L =
3. Three phase active power = ...
4. In a star connected system, the phase difference between line and phase voltage is ....
5. In a 3–phase, delta connected system.
   $I_{ph} = I_L$
6. For the same size, the rating of 3–phase motor will be ...... than the single phase motor.
7. Three phase apparent power = ...
8. In two watt meter method, when the power factor of the load is zero leading or lagging, the two watt meters will give .... reading.
9. In two watt meter method of measuring power in 3–phase network, one watt meter will indicate backward reading if the power factor of the load is less than ....
10. kW = kVA × ....
11. If a 3–phase star connected alternator delivers 60 amp. load current at 0·7 P.F. and at line voltage of 400 V, then
    (a) its phase voltage = ... V.
    (b) its phase current = ... A.
    (c) its kVA rating is = ... kVA.
    (d) its kW load delivered is .... kW
12. By two watt meter method, to determine power factor, the relation used is

$$\tan \phi = \frac{\dots\dots\dots}{W_1 + W_2}$$

13. In a two watt meter method of power measurement in a 3-phase balanced system. The two watt meters will read equally, when the power factor is ....

14. A 20 H.P, 400 V, 3-phase induction motor takes 20 amperes at 0·8 p.f. then,

(a) its kVA = ... (b) its kW ....

15. Wattmeter is an instrument which measures ....

Type–II Mark the most correct answer :

1. In two wattmeter method of measuring power in balanced 3-phase circuit, if the readings of two wattmeters are equal but opposite in sign, then the power factor of the load is

(a) unity (b) 1/2
(c) zero (d) 0·5

2. To transmit the same amount of power over a particular distance at a given voltage, the amount of conducing material required in a 3-phase system to that of single phase system will be

(a) 1·5 times (b) $\sqrt{3}$ times
(c) 3/4th times (d) 0·5 times

3. For the same rating, the size of 3-phase motor to that of a single phase motor will be

(a) More (b) Less
(c) Same (d) None of the above

4. When two wattmeters are used to measure power of a 3-phase balanced circuit and one watt meter reads negative, it means the power factor of the load is

(a) 0 (b) more than 0·5
(c) less than 0·5 (d) 1

5. In a balanced 3-phase, star connected system, the phase difference berween phase voltages and their respective line voltages is

(a) 120° (b) 60°
(c) 45° (d) 30°

6. In a 3-phase, balanced load, the power consumed is given by relation,

(a) $3\ V_L.\ I_L.\ \cos\phi$ (b) $\sqrt{3}\ V_L.\ I_L\ \cos\phi$
(c) $\sqrt{3}\ V_{ph}.\ I_{ph}.\ \cos\phi$ (d) $\sqrt{3}\ V_L.\ I_L.$

7. Three delta connected resistors absorb 30 kW, when connected to a 400 V, 3-phase supply. When they are connected in star across the same supply, the power absorbed will be

(a) 50 kW. (b) 20 kW.
(c) 10 kW. (d) 60 kW.

8. In two watt meter method of measuring power in a balanced 3-phase circuit, if the reading of one watt meter is zero, then the power factor of the load is

(a) unity (b) $\sqrt{3}/2$

9. Three capacitors, each of 150 μF are connected in delta. The value of capacitance in each leg of the equivalent star connected load would be

(a) 450 μF (b) 950 μF

(c) 225 μF (d) 150 μF

10. For power measurement in a 3-phase balanced load, the two wattmeters will read,

(a) $\boxed{W_1 = V_L.I_L. \cos(30 - \phi)}$ and $\boxed{W_2 = V_L.I_L \cos(30 + \phi)}$

(b) $W_1 = \sqrt{3}\ V_L.I_L \cos(30 - \phi)$ and $W_2 = \sqrt{3}\ V_L.I_L \cos(30 + \phi)$

(c) $W_1 = 3\ V_{ph}.I_{ph} \cos\phi$

11. Three identical resistances each of 15Ω are connected in star across a 400 V, 3-phase supply. The value of resistance in each phase of the equivalent delta-connected load would be

(a) 5 Ω (b) 45 Ω

(c) 75 Ω (d) 50 Ω

Type–III True or False Statements.

1. In a 3-phase delta connected system, the apparent power is $\sqrt{3}\ V_L.I_L.\cos\phi$
2. In a 3-phase, 3 wire unbalanced load, power can not be measured by two watt meter method.
3. In a 3-phase, star connected system, $I_{ph} = \sqrt{3}.I_L$
4. kW = kVA × cos ϕ
5. Power of a three phase system is 3 times the phase power.
6. The phase sequence of a 3-phase system is RYB.
7. The power drawn by a pure inductance is zero.
8. The power factor of a pure resistance circuit is zero.
9. Watt meter is an instrument which measures apparent power.
10. In a two wattmeter method of power measurement in a 3-phase balanced system, the two wattmeters will read equally, when the p.f. is unity.

Type–IV Answer the following in brief :–

Q. 1. What are the advantages of 3-phase system over a single phase system ?

Q. 2. How power is measured by two wattmeters in a 3-phase unbalanced load ?

Q. 3. Why fixed coils of dynamometer type wattmeter are connected in series with the load ?

Q. 4. How the power factor of a 3-phase balanced load can be determined using two wattmeters ?

Q. 5. Under what conditions, the two wattmeters read equal and opposite, when connected to measure power.

Q. 6. How the power factor of a 3-phase unbalanced load can be determined using two wattmeters ?

ANSWERS

Type–I

1. 120° electrical, 2. I_{ph}, $\sqrt{3}$, V_{ph}, 3. $\sqrt{3}\ V_L.I_L \cos\phi$ 4. 30°,

5. $\frac{1}{\sqrt{3}}$ 6. more, 7. $\sqrt{3}\,V_L.I_L$,

8. equal and opposite, 9. 0·5, 10. cos ϕ,

11. (a) $\frac{400}{\sqrt{3}}$, (b) 60, (c) 41·5 (d) 29·09,

12. $\sqrt{3}\ (W_1 - W_2)$ 13. unity, 14. (a)13·84, (b) 11·072,

15. average real power

Type–II

1. (c) 2. (c), 3. (b), 4. (c), 5. (d), 6. (b),

7. (c), 8. (c), 9. (a) 10. (a), 11. (b).

Type–III

1. (F), 2. (F), 3. (F), 4. (T), 5. (T), 6. (T),

7. (T), 8. (F), 9. (F) 10. (T).

TEST QUESTIONS

1. What do you mean by polyphase system ?
2. What is a double script notatian ? Explain in details.
3. Three phase system has only two types of connections viz star connection and delta connection. Why ?
4. What do you mean by phase sequence ? How will you check the phase sequence ?
5. What is the importance of phase sequence in 3-phase circuits ?
6. Derive the relation between the line voltage and phase voltage for a balanced 3 phase star connected system.
7. Derive the relation between the line current and phase current for balanced 3 phase delta connected load.
8. Compare star and delta systems of three phase connections
9. Show that total power in a 3-phase star or delta connected circuit is $\sqrt{3}\ V_L\ I_L\ \cos\phi$